

PIONEER JUNIOR COLLEGE

2011 JC2 PRELIMINARY EXAMINATION
HIGHER 1



CHEMISTRY

8872/01

Paper 1

23 September 2011

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

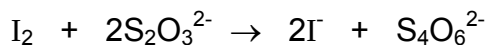
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1** The relative atomic mass of copper, which consists of the isotopes $^{63}_{29}\text{Cu}$ and $^{65}_{29}\text{Cu}$ is 63.5. What is the percentage of $^{63}_{29}\text{Cu}$ atoms in the isotopic mixture?
- A** 0.25%
B 0.75%
C 25%
D 75%
- 2** When 20 cm³ of a hydrocarbon were burnt in 150 cm³ of oxygen (an excess), the final gaseous mixture had a total volume of 110 cm³. On passing through aqueous potassium hydroxide, the volume of gaseous mixture decreased by 60 cm³. All gaseous volumes were measured under room temperature and pressure.
- What is the formula of the hydrocarbon?
- A** C₃H₄
B C₃H₈
C C₆H₁₂
D C₆H₁₄
- 3** Thallium(III) ions can oxidise iodide to iodine.

In an experiment, 25.0 cm³ of a 0.100 mol dm⁻³ solution containing Tl³⁺ ions is added to excess potassium iodide solution. The liberated iodine then reacts with 40.00 cm³ of 0.125 mol dm⁻³ of sodium thiosulfate according to the equation:



What is the oxidation state of Tl after its reaction with potassium iodide?

- A** +2 **B** +1 **C** -1 **D** -2

- 4 The table below contains incomplete information about the two ions X and Y.

Ion	Mass number	Atomic number	Number of neutrons	Number of electrons	Charge of ion
X	40	20	p	q	+2
Y	32	r	16	18	s

Which of the following shows the missing numbers (p, q, r and s) in the table above?

	p	q	r	s
A	20	20	16	-2
B	20	18	16	-2
C	20	20	20	+2
D	20	18	20	+2

- 5 What is the electronic configuration of Cr^{3+} ? [Ar] represents $1s^2 2s^2 2p^6 3s^2 3p^6$.
- A [Ar] $3d^1 4s^2$
 B [Ar] $3d^2 4s^2$
 C [Ar] $3d^3 4s^0$
 D [Ar] $3d^4 4s^0$
- 6 In which of the following pairs does the first ion have a larger bond angle than the second?
- A ClO_3^- BrO_3^-
 B PCl_6^- PCl_4^+
 C SO_3^{2-} SO_4^{2-}
 D NO_2^- NO_2^+
- 7 Which of the following statements about the properties associated with ionic and covalent bonds is correct?
- A The only covalent compounds with high melting points are those in which hydrogen bonds occur.
 B Any covalent compounds that contain both nitrogen and hydrogen in its molecules form hydrogen bonds.
 C Ionic bonds and covalent bonds cannot occur in the same compound.
 D Ionic compounds differ from metals in that ionic compounds do not conduct electricity in the solid state.

- 8 The standard enthalpy changes of formation of carbon dioxide and water are -394 kJ mol^{-1} and -286 kJ mol^{-1} respectively. If the standard enthalpy change of combustion of ethanol is $-1367 \text{ kJ mol}^{-1}$, what is its enthalpy change of formation?

A -279 kJ mol^{-1}
 B -687 kJ mol^{-1}
 C $+279 \text{ kJ mol}^{-1}$
 D $+687 \text{ kJ mol}^{-1}$

- 9 Enthalpy change of solution of sodium chloride is defined as the heat absorbed when one mole of sodium chloride dissolves in water to form an infinitely dilute solution.

When 11.7 g of sodium chloride (M_r of NaCl = 58.5) dissolves in 100 g of water, temperature of the solution falls by 2.4°C . Given the specific heat capacity of solution to be $4.2 \text{ J g}^{-1} \text{ K}^{-1}$, what is the enthalpy change of solution of sodium chloride?

A $+\frac{(100+11.7) \times 4.2 \times (2.4+273) \times 58.5}{1000 \times 11.7} \text{ kJ mol}^{-1}$
 B $+\frac{(100+11.7) \times 4.2 \times 2.4 \times 58.5}{1000 \times 11.7} \text{ kJ mol}^{-1}$
 C $+\frac{100 \times 4.2 \times (2.4+273) \times 58.5}{1000 \times 11.7} \text{ kJ mol}^{-1}$
 D $+\frac{100 \times 4.2 \times 2.4 \times 58.5}{1000 \times 11.7} \text{ kJ mol}^{-1}$

- 10 Which of the following statements about the forward and reverse reactions, $\text{P} + \text{Q} \rightleftharpoons \text{R} + \text{S}$, is correct when the system is in equilibrium?

A $[\text{P}][\text{Q}] = [\text{R}][\text{S}]$
 B The rate constant for the forward reaction equals to the rate constant for the reverse reaction.
 C The rates of the forward and reverse reactions are equal.
 D The rates of the forward and reverse reactions are equal to zero.

- 11** Phosgene, COCl_2 , is a poisonous gas used as a chemical weapon in World War I. At high temperature, it can decompose to give carbon monoxide and chlorine, according to the following equation.



In an experiment, 4 mol of phosgene was put into a 2 dm^3 container and heated. The equilibrium mixture was found to contain 0.8 mol of chlorine. What is the numerical value of the equilibrium constant at the temperature of the experiment?

- A** 0.10 **B** 1.3 **C** 0.20 **D** 0.27

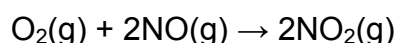
- 12** The table below shows the values of the ionic product of water, K_w , at two different temperatures.

Temperature / $^{\circ}\text{C}$	$K_w / \text{mol}^2 \text{ dm}^{-6}$
25	1.00×10^{-14}
62	1.00×10^{-13}

Which of the following is correct for pure water at 62°C ?

- A** $[\text{H}^+] > [\text{OH}^-]$
B $\text{pOH} = \text{pH} < 7$
C $\text{pH} = 7$
D $\text{pH} = 14 - \text{pOH}$

- 13** Oxygen reacts with nitrogen monoxide according to the equation shown.



To investigate the effects of concentrations on the rate of the reaction, the following experimental results were obtained.

Expt	$[\text{O}_2] / \text{mol dm}^{-3}$	$[\text{NO}] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	1.0	1.0	0.0007
2	2.0	1.0	0.0014
3	0.5	2.0	0.0007
4	2.0	1.5	y

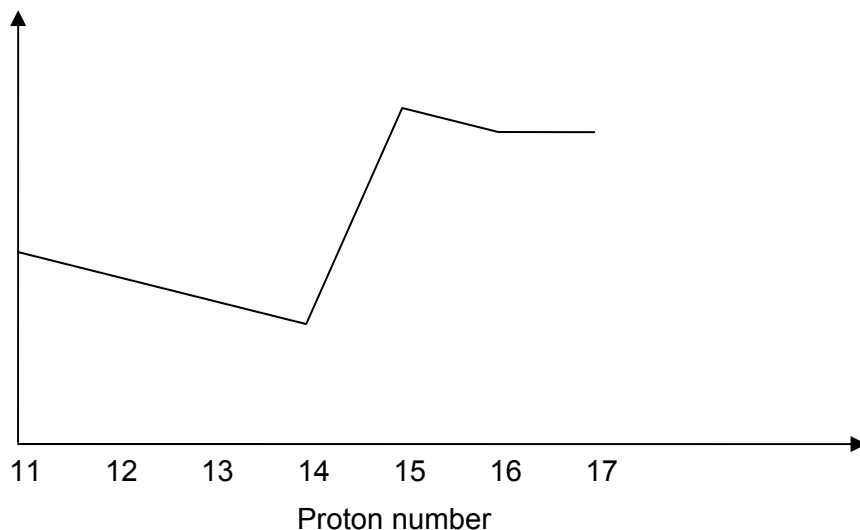
What is the value of **y**?

- A** 0.0007
B 0.0011
C 0.0014
D 0.0021

- 14 Which of the following elements has an oxide with a giant structure and a chloride which does **not** hydrolyse?

A sodium B aluminium C silicon D phosphorus

- 15 The graph shows how a property of the elements Na to Cl varies with proton number.

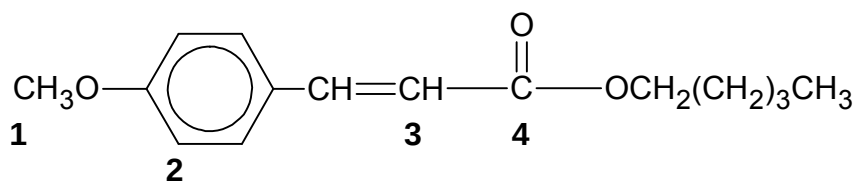


What is the property?

- A melting point
B ionic radius
C first ionisation energy
D electronegativity
- 16 How many different alkenes, including geometric isomers, could be produced by the removal of HBr from $\text{C}_6\text{H}_5\text{CHBrCH}_2\text{CH}_3$?

A 1 B 2 C 3 D 4

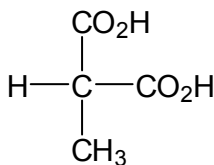
- 17 Pentyl 4-methoxycinnamate is used in various sun-tan creams to absorb excessive ultraviolet radiation and stop the skin burning.



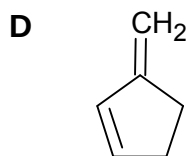
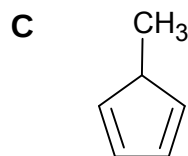
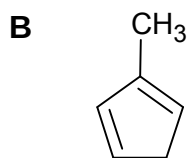
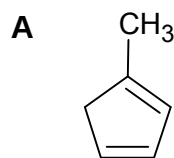
Pentyl-4-methoxycinnamate

Which statement about pentyl-4-methoxycinnamate is correct?

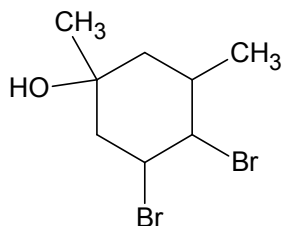
- A The carbon atom **1** is sp^2 hybridised.
 - B The carbon atom **3** is sp^3 hybridised.
 - C The carbon atom **2** has trigonal planar geometry.
 - D The carbon atom **4** has T-shape geometry.
- 18 A hydrocarbon, C_6H_8 , on heating with acidified potassium manganate(VII) produces the structure below as the only organic product.



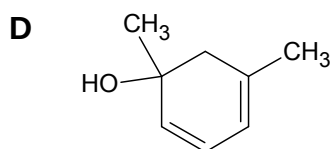
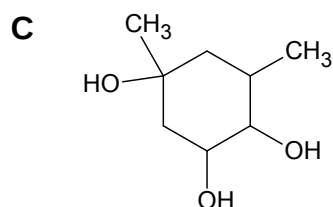
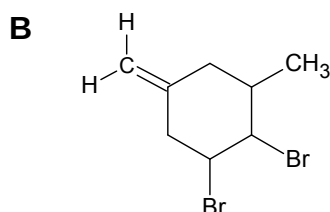
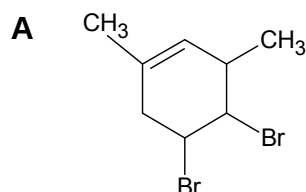
What could be a possible structure of the hydrocarbon?



- 19 The following compound was heated with ethanolic sodium hydroxide.



Which of the following represents the structure of the organic product?



- 20 Ozone depletion potential (ODP) is a measure of the effectiveness of chlorofluoroalkanes in destroying stratospheric ozone.

In which sequence are compounds listed in increasing order of their ODPs?

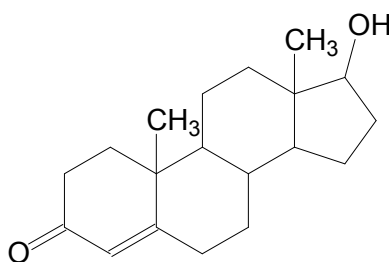
- A** $\text{CH}_3\text{CCl}_2\text{F} < \text{CCl}_2\text{FCClF}_2 < \text{CHClF}_2$
B $\text{CCl}_2\text{FCClF}_2 < \text{CHClF}_2 < \text{CH}_3\text{CCl}_2\text{F}$
C $\text{CHClF}_2 < \text{CCl}_2\text{FCClF}_2 < \text{CH}_3\text{CCl}_2\text{F}$
D $\text{CHClF}_2 < \text{CH}_3\text{CCl}_2\text{F} < \text{CCl}_2\text{FCClF}_2$

- 21 An alcohol was treated with hot acidified potassium dichromate(VI). The separated product gave an orange precipitate with 2,4-dinitrophenylhydrazine and a yellow precipitate with alkaline aqueous iodine.

Which alcohol was used?

- A** $\text{CH}_3\text{C}(\text{CH}_3)_2\text{OH}$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
C $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$
D $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

- 22 Testosterone is a steroid hormone found in mammals, reptiles, birds and other vertebrates.



testosterone

Testosterone is first reacted with hydrogen in the presence of a platinum catalyst, and the product is then warmed with acidified KMnO_4 .

Given that no carbon-carbon σ -bond is broken in this process, how many double bonds will there be in the structure of the final product?

- A 0 B 1 C 2 D 3

- 23 Compound **A** was subjected to the following reactions in the sequence given.

Step 1: heat with ethanolic NaCN

Step 2: heat with dilute HCl

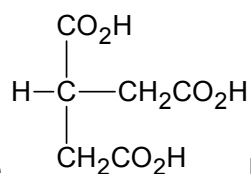
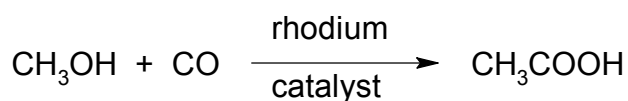
Step 3: heat with $\text{CH}_3\text{CH}_2\text{OH}$ in the presence of concentrated H_2SO_4

The final product obtained was $\text{CH}_3\text{CH}_2\text{OCOCH}_2\text{CH}_3$.

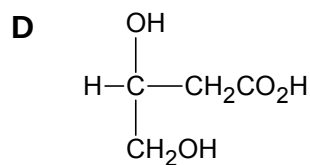
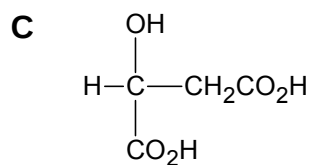
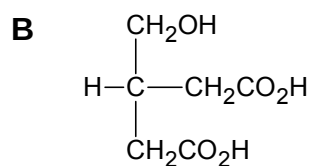
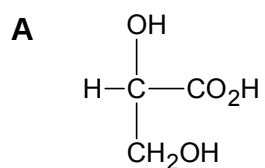
What could compound **A** be?

- A $\text{CH}_3\text{CH}_2\text{Br}$
 B $\text{CH}_3\text{CH}_2\text{OH}$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
 D $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- 24 Which compound is produced when $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$ reacts with hot aqueous sodium hydroxide?
- A $\text{CH}_3\text{CH}_2\text{OH}$
 B $\text{CH}_3\text{CO}_2^-\text{Na}^+$
 C $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
 D $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}^-\text{Na}^+$
- 25 One industrial preparation of ethanoic acid is the direct carbonylation of methanol using a rhodium catalyst.



Which compound could be expected to produce by this method?



Section B

For each question in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Gaseous particle **X** has a proton number n and a charge of $+1$.

Gaseous particle **Y** has a proton number of $(n + 1)$ and is isoelectronic with **X**.

- 1** **X** has a larger radius than **Y**.
- 2** **X** requires less energy than **Y** when a further electron is removed from each particle.
- 3** The magnitude of lattice energy for the oxide of **X** is greater than that of **Y**.

27 Which of the following are correct descriptions of a strong monoprotic acid?

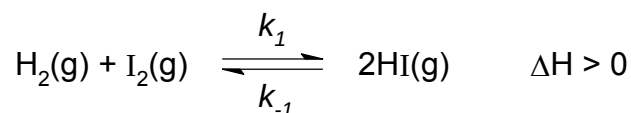
- 1** It has a low pK_a .
- 2** It has a relatively high electrical conductivity in dilute solution.
- 3** It requires a greater volume of alkali for complete neutralisation compared to a weak monoprotic acid of the same concentration.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

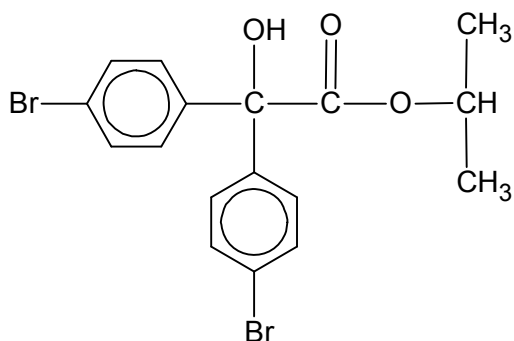
- 28** Hydrogen and iodine exist in equilibrium with hydrogen iodide at a constant temperature.



Which of the following changes will cause both of the rate constants of the forward, k_1 for the forward reaction and k_{-1} for the reverse reaction as well as the equilibrium constant, K_c to increase?

- 1** heating the mixture
- 2** using a catalyst
- 3** increasing the concentrations of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$

- 29** *Acarol* is sold as an insecticide for use on fruits and vegetables.



Acarol

Which statements about *Acarol* are correct?

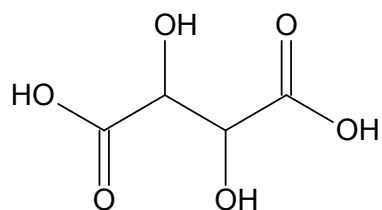
- 1** It reacts with hot ethanolic potassium cyanide
- 2** It gives a yellow precipitate on treatment with hot alkaline aqueous iodine.
- 3** It turns orange acidified potassium dichromate(VI) green on heating.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

30 Tartaric acid, one of the main acids found in wine, has the following formula.



tartaric acid

Which of the following statements are correct?

- 1** 1 mol of tartaric acid reacts with excess Na(s) to give 2 mol of H₂(g).
- 2** 1 mol of tartaric acid reacts with an excess of PCl₅(s) to form 4 mol of HCl(g).
- 3** 1 mol of tartaric acid reacts with completely with 4 mol of HBr(g).

End of Paper

Pioneer Junior College
2011 JC2 Preliminary Examination
H1 Chemistry (8872)
Answers

1	D	11	A	21	D
2	B	12	B	22	C
3	B	13	D	23	A
4	B	14	A	24	B
5	C	15	B	25	D
6	A	16	B	26	B
7	D	17	C	27	B
8	A	18	C	28	D
9	D	19	D	29	C
10	C	20	D	30	B